

Dear Editors,

The authors described this book to me and asked whether I would take part in it. Having seen the plan, I write to support the proposal and to confirm that I intend to contribute.

I work on the research and development of electrical machines at ABB, where high-speed machines are products rather than demonstrations. What strikes me about this plan is that it treats manufacture, tolerances and qualification as part of the design rather than as what happens afterwards. In our experience that is the right way round. A design that looks excellent on paper may still be commercially unviable and cannot be manufactured consistently or economically, because the tolerances that decide the outcome at these speeds are tighter than a designer is accustomed to specifying, and because the assembly sequence constrains the geometry before the electromagnetic design ever sees it. Very little of this appears in print.

That is also the reason I think the book is needed. When we bring a new engineer into high-speed work, there is no text to give them. They learn from internal documents, from drawings, and from colleagues who will in due course retire. The research literature is plentiful, but it is organised one paper to one phenomenon, whereas a product requires all of the phenomena settled at once and against each other. A book that sets out the design sequence, states which constraint binds first, and carries the example machines through from specification to measured performance would be of immediate use to us, and I would expect the same at other manufacturers.

Subject to the publishing contract, I intend to contribute the section on induction machines, which is the topology behind a large part of industrial high-speed drive practice and which is better described by someone who builds them than from the literature; the sections on manufacture and tolerance control, and on qualification against international standards, in the industrial case-study chapter, and subject to my employer's agreement, photographs and material from machines we have built. I have worked on the research and development of electrical machines at ABB over 30 years, including different kinds of motors and generators.

Yours sincerely,

A handwritten signature in blue ink, appearing to read 'J. Ikäheimo', with a small mark at the end.

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